Example 1

```
supplement_heart_rate |> anovas(outcome = resting_heart_rate_bpm,
                                A = treatment,
                                B = pony_type,
                                interaction = TRUE)
```

```
Two-way ANOVA (with interaction)

Source               df         SS        MS       F  p-value
──────────────────────────────────────────────────────────────
treatment             1   215.2813  215.2813  15.409  < 0.001
pony_type             2  1383.6803  691.8401  49.519  < 0.001
treatment:pony_type   2   636.6508  318.3254  22.785  < 0.001
Residuals            66   922.0942   13.9711
──────────────────────────────────────────────────────────────


Hypotheses (treatment:pony_type):
H₀: There is no interaction between treatment and pony_type
H₁: There is an interaction between treatment and pony_type
Test statistic: F(2, 66) = 22.785
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

```
supplement_heart_rate |> posthoc(outcome = resting_heart_rate_bpm,
                                A = treatment,
                                B = pony_type,
                                interaction = TRUE)
```

```
Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA (resting_heart_rate_bpm ~ treatment * pony_type)

Main effect: treatment
Comparison             Difference  p-value
──────────────────────────────────────────────
Supplement vs Placebo     -3.4583  < 0.001  *
──────────────────────────────────────────────

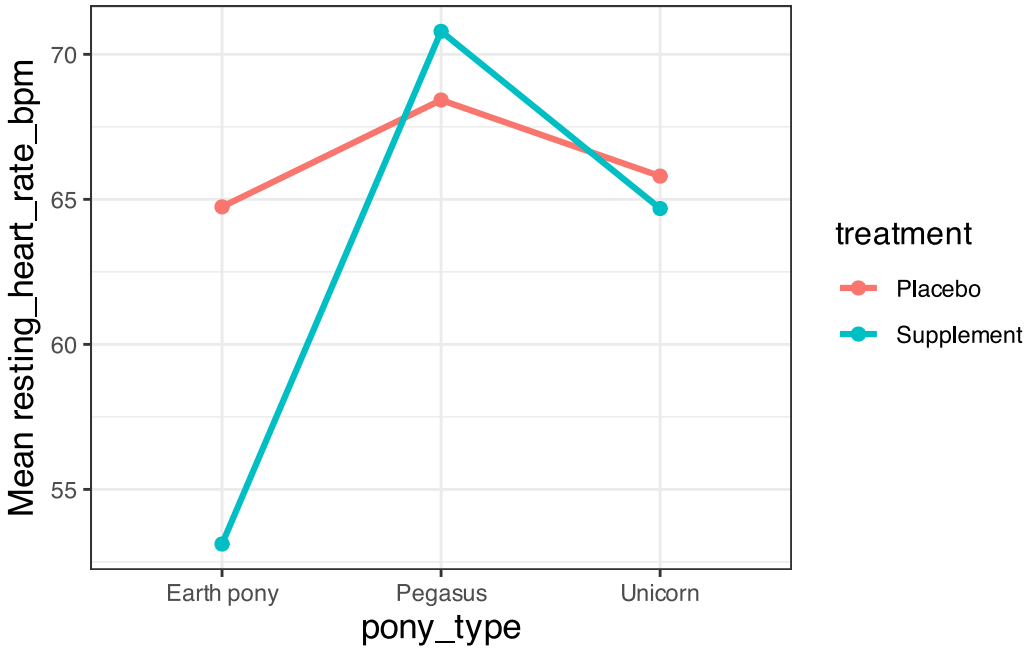
Main effect: pony_type
Comparison             Difference  p-value
──────────────────────────────────────────────
Pegasus vs Earth pony     10.6792  < 0.001  *
Unicorn vs Earth pony      6.3125  < 0.001  *
Unicorn vs Pegasus        -4.3667  < 0.001  *
```

```
──────────────────────────────────────────────────

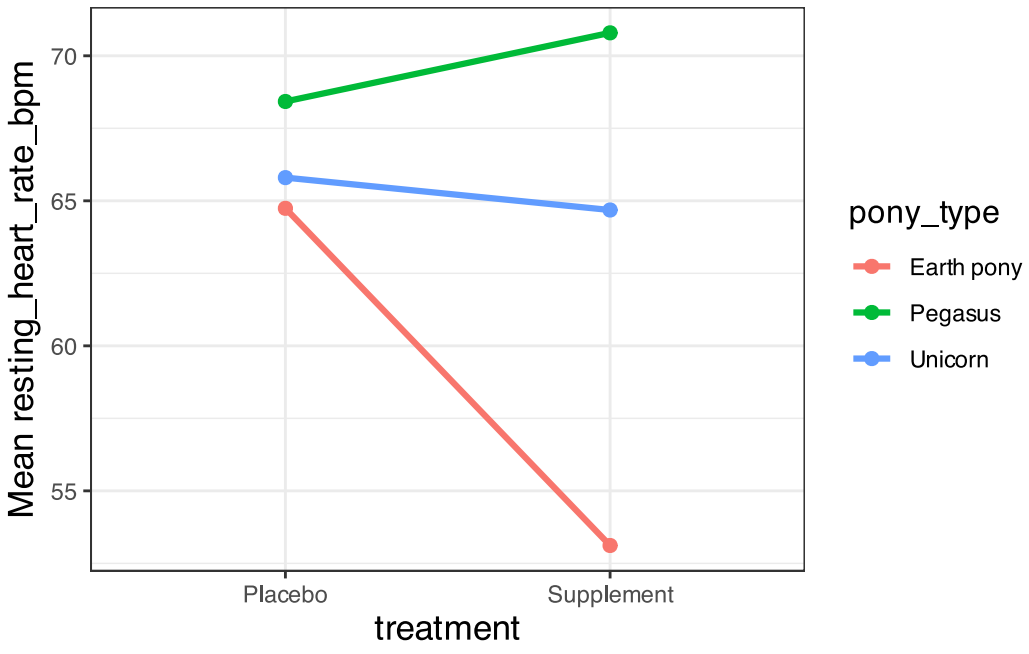

Interaction: treatment × pony_type
Comparison                                      Difference  p-value
──────────────────────────────────────────────────────────────────
Supplement:Earth pony vs Placebo:Earth pony    -11.6250  < 0.001  *
Placebo:Pegasus vs Placebo:Earth pony            3.6833    0.167
Supplement:Pegasus vs Placebo:Earth pony         6.0500    0.002  *
Placebo:Unicorn vs Placebo:Earth pony            1.0583    0.982
Supplement:Unicorn vs Placebo:Earth pony        -0.0583    1.000
Placebo:Pegasus vs Supplement:Earth pony        15.3083  < 0.001  *
Supplement:Pegasus vs Supplement:Earth pony     17.6750  < 0.001  *
Placebo:Unicorn vs Supplement:Earth pony        12.6833  < 0.001  *
Supplement:Unicorn vs Supplement:Earth pony     11.5667  < 0.001  *
Supplement:Pegasus vs Placebo:Pegasus            2.3667    0.633
Placebo:Unicorn vs Placebo:Pegasus              -2.6250    0.524
Supplement:Unicorn vs Placebo:Pegasus           -3.7417    0.154
Placebo:Unicorn vs Supplement:Pegasus           -4.9917    0.020  *
Supplement:Unicorn vs Supplement:Pegasus        -6.1083    0.002  *
Supplement:Unicorn vs Placebo:Unicorn           -1.1167    0.977
──────────────────────────────────────────────────────────────────


* p < α = 0.050
```

```
supplement_heart_rate |> profile_plot(outcome = resting_heart_rate_bpm,
                                      lines = treatment,
                                      x = pony_type)
```




```
supplement_heart_rate |> profile_plot(outcome = resting_heart_rate_bpm,
                                      lines = pony_type,
                                      x = treatment)
```



Example 2

```
exercise_recovery |> anovas(outcome = recovery_time_min,
                            A = treatment,
                            B = pony_type,
                            interaction = FALSE)
```

```
Two-way ANOVA (no interaction)

Source      df         SS         MS       F  p-value
──────────────────────────────────────────────────────
treatment    2  2909.4913  1454.7456  80.254  < 0.001
pony_type    2    23.1424    11.5712   0.638    0.530
Residuals  103  1867.0562    18.1268
──────────────────────────────────────────────────────


Hypotheses (treatment):
H₀: All means across levels of treatment are equal
H₁: At least one mean across levels of treatment differs
Test statistic: F(2, 103) = 80.254
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)

Hypotheses (pony_type):
H₀: All means across levels of pony_type are equal
H₁: At least one mean across levels of pony_type differs
Test statistic: F(2, 103) = 0.638
p-value: 0.530
Conclusion: Fail to reject the null hypothesis (p = 0.530 ≥ α = 0.050)
```

```
exercise_recovery |> posthoc(outcome = recovery_time_min,
                            A = treatment,
                            B = pony_type,
                            interaction = FALSE)
```

```
Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA (recovery_time_min ~ treatment + pony_type)

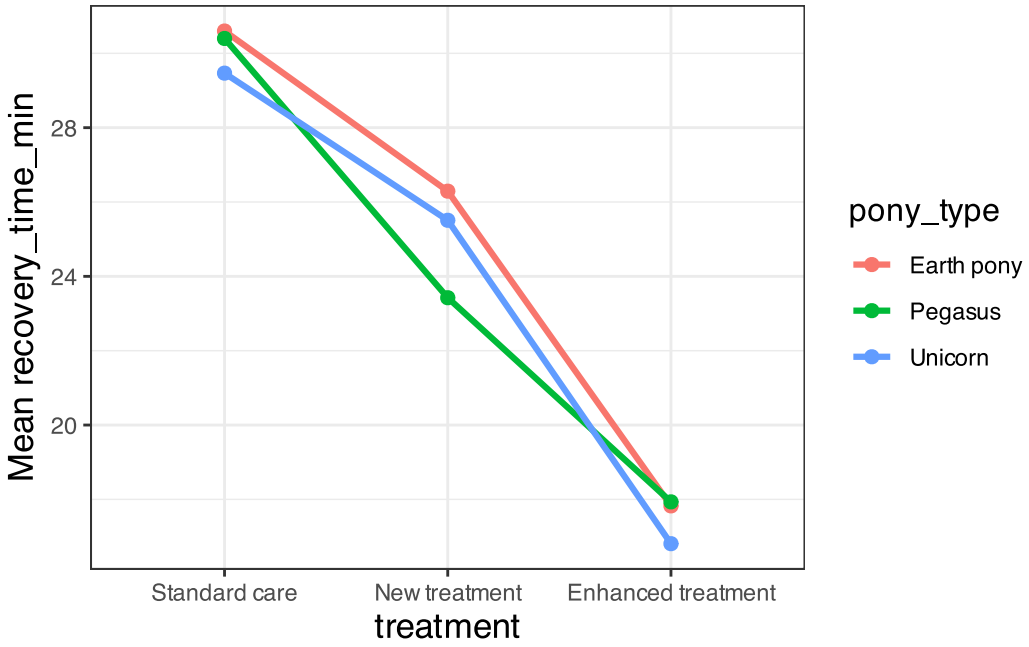
Main effect: treatment
Comparison                          Difference  p-value
──────────────────────────────────────────────────────────
New treatment vs Standard care         -5.0806  < 0.001  *
Enhanced treatment vs Standard care   -12.6333  < 0.001  *
Enhanced treatment vs New treatment    -7.5528  < 0.001  *
```

```
──────────────────────────────────────────────────────────────

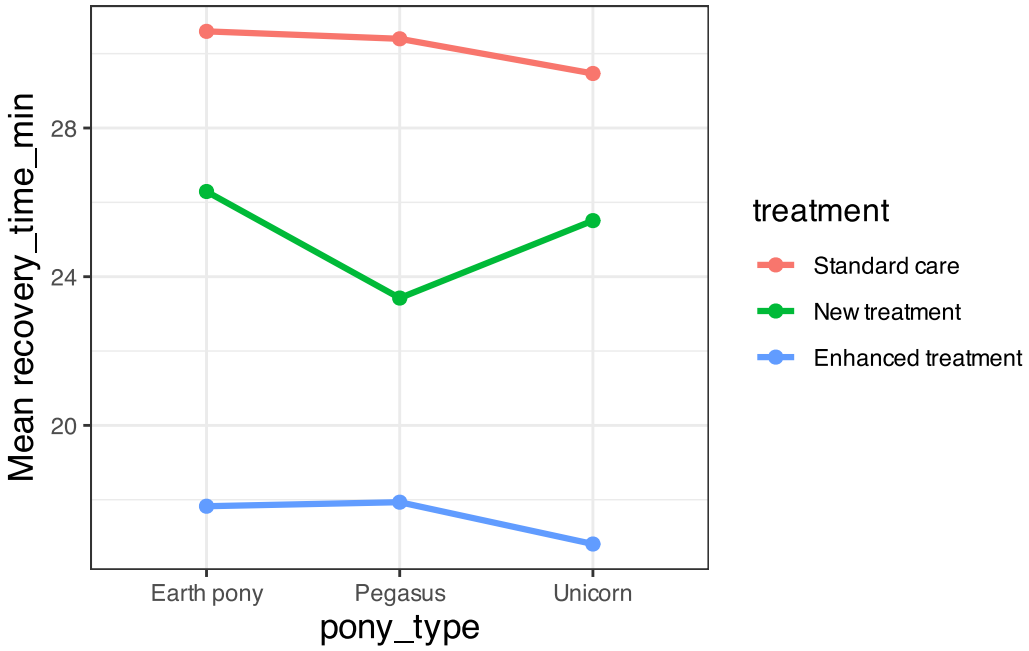
Main effect: pony_type
Comparison             Difference  p-value
──────────────────────────────────────────────────
Pegasus vs Earth pony     -0.9861    0.589
Unicorn vs Earth pony     -0.9778    0.595
Unicorn vs Pegasus         0.0083    1.000
──────────────────────────────────────────────────

* p < α = 0.050
```

```
exercise_recovery |> profile_plot(outcome = recovery_time_min,
                                  xaxis = treatment,
                                  lines = pony_type)
```




```
exercise_recovery |> profile_plot(outcome = recovery_time_min,
                                  xaxis = pony_type,
                                  lines = treatment)
```


Mean recovery_time_min
28
24
20
Earth pony
Pegasus
Unicorn
pony_type
treatment
Standard care
New treatment
Enhanced treatment

Let's Practice 1

Applejack and the Ponyville clinic are studying whether a high-fiber feed changes postmeal blood glucose. Within each pony type, adult ponies are randomly assigned to Standard feed or High-fiber feed for two weeks. After a standardized meal, glucose is measured once at the same pre-specified time for every pony.

```
glucose |> anovas(outcome = postmeal_glucose_mg_dl,
                  A = treatment,
                  B = pony_type)
```

```
Two-way ANOVA (with interaction)

Source               df         SS         MS       F  p-value
──────────────────────────────────────────────────────────────────
treatment             1   531.4410   531.4410  13.231  < 0.001
pony_type             2  2111.6527  1055.8263  26.286  < 0.001
treatment:pony_type   2  2635.6647  1317.8323  32.809  < 0.001
Residuals            84  3374.0227    40.1669
──────────────────────────────────────────────────────────────────


Hypotheses (treatment:pony_type):
H₀: There is no interaction between treatment and pony_type
H₁: There is an interaction between treatment and pony_type
Test statistic: F(2, 84) = 32.809
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

```
glucose |> posthoc(outcome = postmeal_glucose_mg_dl,
                   A = treatment,
                   B = pony_type)
```

```
Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA (postmeal_glucose_mg_dl ~ treatment * pony_type)

Main effect: treatment
Comparison                          Difference  p-value
──────────────────────────────────────────────────────────
Standard feed vs High vs fiber feed      4.8600  < 0.001  *
──────────────────────────────────────────────────────────

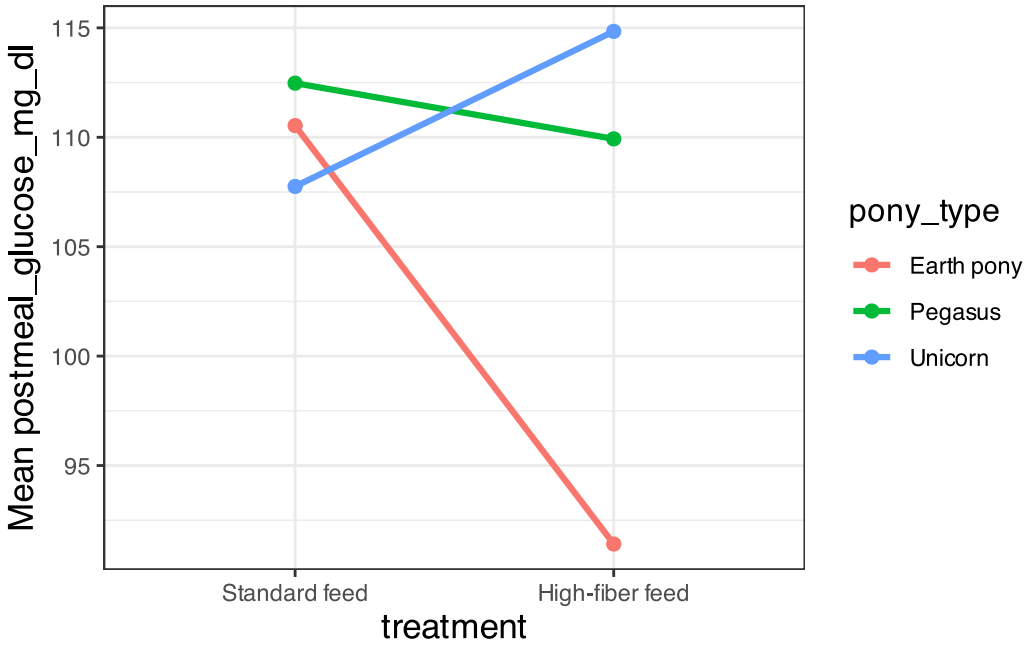

Main effect: pony_type
Comparison             Difference  p-value
──────────────────────────────────────────
```

```
Pegasus vs Earth pony     10.2267  < 0.001  *
Unicorn vs Earth pony     10.3233  < 0.001  *
Unicorn vs Pegasus         0.0967    0.998
──────────────────────────────────────────────

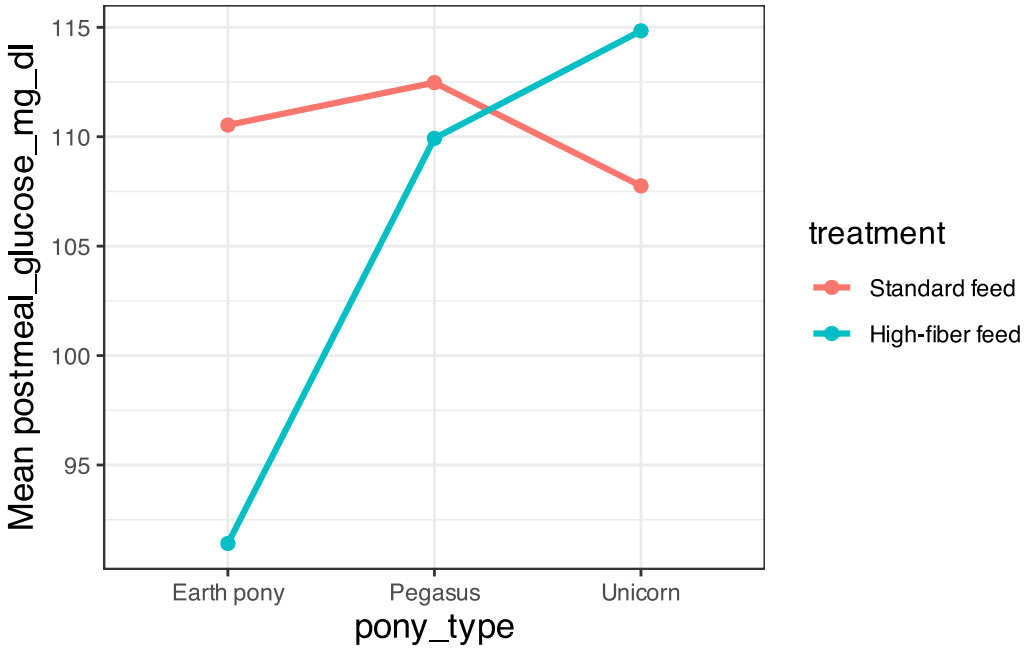

Interaction: treatment × pony_type
Comparison                                                    Difference  p-value
──────────────────────────────────────────────────────────────────────────────────
Standard feed:Earth pony vs High vs fiber feed:Earth pony       19.1200  < 0.001
*
High vs fiber feed:Pegasus vs High vs fiber feed:Earth pony     18.5133  < 0.001
*
Standard feed:Pegasus vs High vs fiber feed:Earth pony          21.0600  < 0.001
*
High vs fiber feed:Unicorn vs High vs fiber feed:Earth pony     23.4267  < 0.001
*
Standard feed:Unicorn vs High vs fiber feed:Earth pony          16.3400  < 0.001
*
High vs fiber feed:Pegasus vs Standard feed:Earth pony          -0.6067    1.000
Standard feed:Pegasus vs Standard feed:Earth pony                1.9400    0.959
High vs fiber feed:Unicorn vs Standard feed:Earth pony           4.3067    0.433
Standard feed:Unicorn vs Standard feed:Earth pony               -2.7800    0.835
Standard feed:Pegasus vs High vs fiber feed:Pegasus              2.5467    0.880
High vs fiber feed:Unicorn vs High vs fiber feed:Pegasus         4.9133    0.286
Standard feed:Unicorn vs High vs fiber feed:Pegasus             -2.1733    0.935
High vs fiber feed:Unicorn vs Standard feed:Pegasus              2.3667    0.909
Standard feed:Unicorn vs Standard feed:Pegasus                  -4.7200    0.329
Standard feed:Unicorn vs High vs fiber feed:Unicorn             -7.0867    0.034
*
──────────────────────────────────────────────────────────────────────────────────


* p < α = 0.050
```

```
glucose |> profile_plot(outcome = postmeal_glucose_mg_dl,
                        xaxis = treatment,
                        lines = pony_type)
```




```
glucose |> profile_plot(outcome = postmeal_glucose_mg_dl,
                        xaxis = pony_type,
                        lines = treatment)
```



Let's Practice 2

Fluttershy is testing a fictional anti-inflammatory treatment in adult ponies with the same mild inflammatory condition. Within each pony type, ponies are randomly assigned to Placebo, Low dose, or High dose. A blood inflammation marker is measured once after seven days.

```
inflammation |> anovas(outcome = inflammation_marker_mg_l,
                        A = treatment,
                        B = pony_type)
```

```
Two-way ANOVA (with interaction)

Source                 df        SS        MS       F  p-value
──────────────────────────────────────────────────────────────
treatment               2  384.4924  192.2462  86.419  < 0.001
pony_type               2  366.1000  183.0500  82.286  < 0.001
treatment:pony_type     4  303.1356   75.7839  34.067  < 0.001
Residuals             126  280.2960    2.2246
──────────────────────────────────────────────────────────────


Hypotheses (treatment:pony_type):
H₀: There is no interaction between treatment and pony_type
H₁: There is an interaction between treatment and pony_type
Test statistic: F(4, 126) = 34.067
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

```
inflammation |> posthoc(outcome = inflammation_marker_mg_l,
                         A = treatment,
                         B = pony_type)
```

```
Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA (inflammation_marker_mg_l ~ treatment * pony_type)

Main effect: treatment
Comparison             Difference  p-value
──────────────────────────────────────────
Low dose vs High dose      0.9378    0.010  *
Placebo vs High dose       3.9556  < 0.001  *
Placebo vs Low dose        3.0178  < 0.001  *
──────────────────────────────────────────

Main effect: pony_type
Comparison             Difference  p-value
```

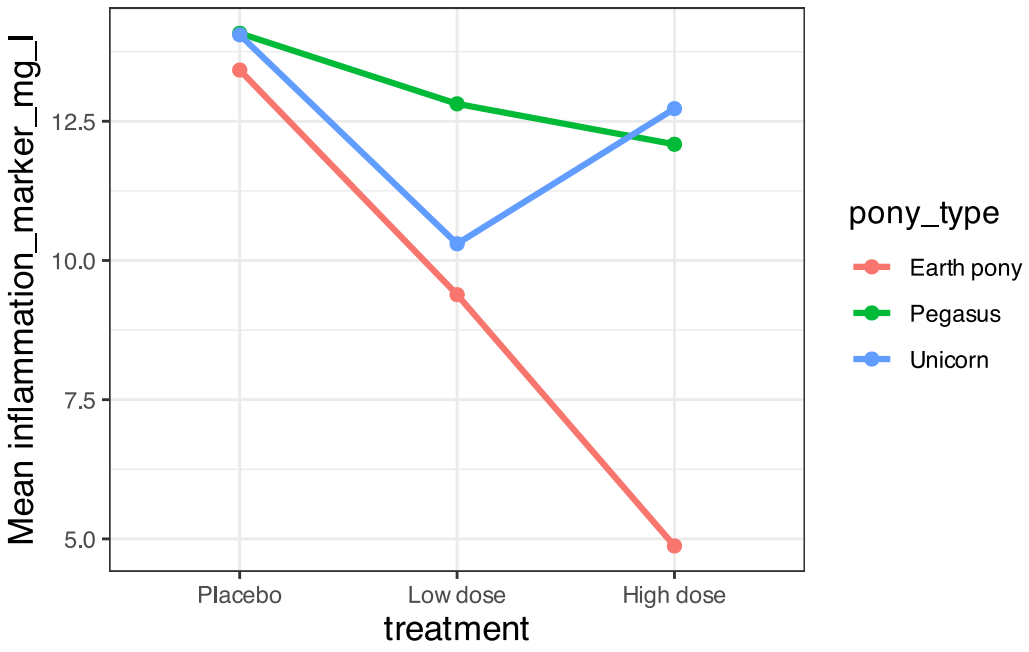
```
──────────────────────────────────────────────
Pegasus vs Earth pony      3.7667  < 0.001  *
Unicorn vs Earth pony      3.1333  < 0.001  *
Unicorn vs Pegasus        -0.6333    0.113
──────────────────────────────────────────────
```

Interaction: treatment × pony_type

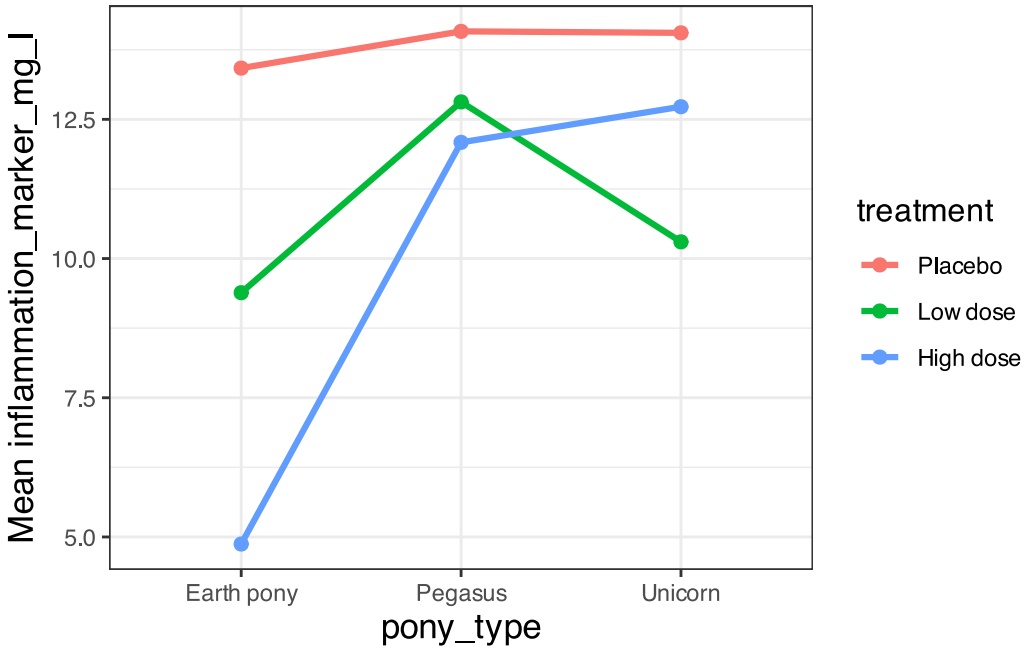
Comparison	Difference	p-value	
Low dose:Earth pony vs High dose:Earth pony	4.5133	< 0.001	*
Placebo:Earth pony vs High dose:Earth pony	8.5467	< 0.001	*
High dose:Pegasus vs High dose:Earth pony	7.2133	< 0.001	*
Low dose:Pegasus vs High dose:Earth pony	7.9400	< 0.001	*
Placebo:Pegasus vs High dose:Earth pony	9.2067	< 0.001	*
High dose:Unicorn vs High dose:Earth pony	7.8533	< 0.001	*
Low dose:Unicorn vs High dose:Earth pony	5.4267	< 0.001	*
Placebo:Unicorn vs High dose:Earth pony	9.1800	< 0.001	*
Placebo:Earth pony vs Low dose:Earth pony	4.0333	< 0.001	*
High dose:Pegasus vs Low dose:Earth pony	2.7000	< 0.001	*
Low dose:Pegasus vs Low dose:Earth pony	3.4267	< 0.001	*
Placebo:Pegasus vs Low dose:Earth pony	4.6933	< 0.001	*
High dose:Unicorn vs Low dose:Earth pony	3.3400	< 0.001	*
Low dose:Unicorn vs Low dose:Earth pony	0.9133	0.759	
Placebo:Unicorn vs Low dose:Earth pony	4.6667	< 0.001	*
High dose:Pegasus vs Placebo:Earth pony	-1.3333	0.268	
Low dose:Pegasus vs Placebo:Earth pony	-0.6067	0.971	
Placebo:Pegasus vs Placebo:Earth pony	0.6600	0.953	
High dose:Unicorn vs Placebo:Earth pony	-0.6933	0.937	
Low dose:Unicorn vs Placebo:Earth pony	-3.1200	< 0.001	*
Placebo:Unicorn vs Placebo:Earth pony	0.6333	0.963	
Low dose:Pegasus vs High dose:Pegasus	0.7267	0.919	
Placebo:Pegasus vs High dose:Pegasus	1.9933	0.011	*
High dose:Unicorn vs High dose:Pegasus	0.6400	0.960	
Low dose:Unicorn vs High dose:Pegasus	-1.7867	0.035	*
Placebo:Unicorn vs High dose:Pegasus	1.9667	0.013	*
Placebo:Pegasus vs Low dose:Pegasus	1.2667	0.335	
High dose:Unicorn vs Low dose:Pegasus	-0.0867	1.000	
Low dose:Unicorn vs Low dose:Pegasus	-2.5133	< 0.001	*
Placebo:Unicorn vs Low dose:Pegasus	1.2400	0.364	
High dose:Unicorn vs Placebo:Pegasus	-1.3533	0.249	
Low dose:Unicorn vs Placebo:Pegasus	-3.7800	< 0.001	*
Placebo:Unicorn vs Placebo:Pegasus	-0.0267	1.000	
Low dose:Unicorn vs High dose:Unicorn	-2.4267	< 0.001	*
Placebo:Unicorn vs High dose:Unicorn	1.3267	0.274	
Placebo:Unicorn vs Low dose:Unicorn	3.7533	< 0.001	*

* p < α = 0.050

```
inflammation |> profile_plot(outcome = inflammation_marker_mg_l,
                              xaxis = treatment,
                              lines = pony_type)
```




```
inflammation |> profile_plot(outcome = inflammation_marker_mg_l,
                              lines = treatment,
                              xaxis = pony_type)
```


Mean inflammation_marker_mg_l
12.5
10.0
7.5
5.0
Earth pony
Pegasus
Unicorn
pony_type
treatment
Placebo
Low dose
High dose

Let's Practice 3

Nurse Redheart compares dressings in adult ponies receiving care for comparable minor wounds. Within each pony type, ponies are randomly assigned to Standard dressing, Hydrogel, or Enhanced hydrogel. The clinic records the time from first dressing to complete wound closure using a common clinical criterion. Each pony contributes one wound.

```
wound_healing |> anovas(outcome = wound_closure_days,
                        A = treatment,
                        B = pony_type,
                        interaction = FALSE)
```

```
Two-way ANOVA (no interaction)

Source      df         SS        MS        F  p-value
──────────────────────────────────────────────────────
treatment    2  1306.4801  653.2401  244.050  < 0.001
pony_type    2   944.2704  472.1352  176.390  < 0.001
Residuals  130   347.9661    2.6767
──────────────────────────────────────────────────────


Hypotheses (treatment):
H₀: All means across levels of treatment are equal
H₁: At least one mean across levels of treatment differs
Test statistic: F(2, 130) = 244.05
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)

Hypotheses (pony_type):
H₀: All means across levels of pony_type are equal
H₁: At least one mean across levels of pony_type differs
Test statistic: F(2, 130) = 176.39
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

```
wound_healing |> posthoc(outcome = wound_closure_days,
                         A = treatment,
                         B = pony_type,
                         interaction = FALSE)
```

```
Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA (wound_closure_days ~ treatment + pony_type)

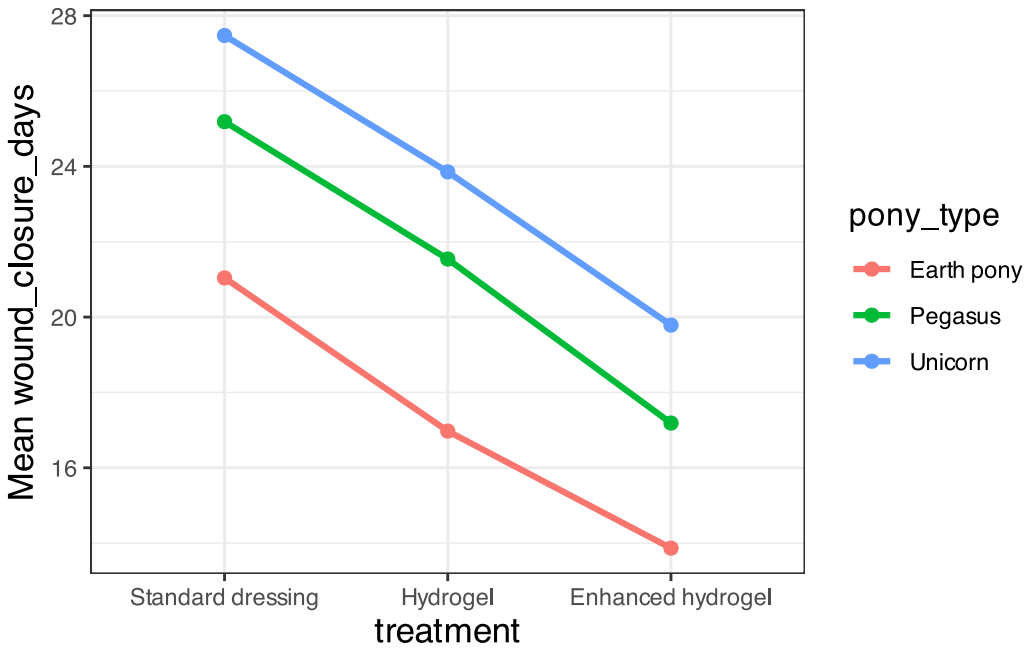
Main effect: treatment
```

```
Comparison                                  Difference  p-value
──────────────────────────────────────────────────────────────────
Hydrogel vs Enhanced hydrogel                   3.8422  < 0.001  *
Standard dressing vs Enhanced hydrogel          7.6200  < 0.001  *
Standard dressing vs Hydrogel                   3.7778  < 0.001  *
──────────────────────────────────────────────────────────────────

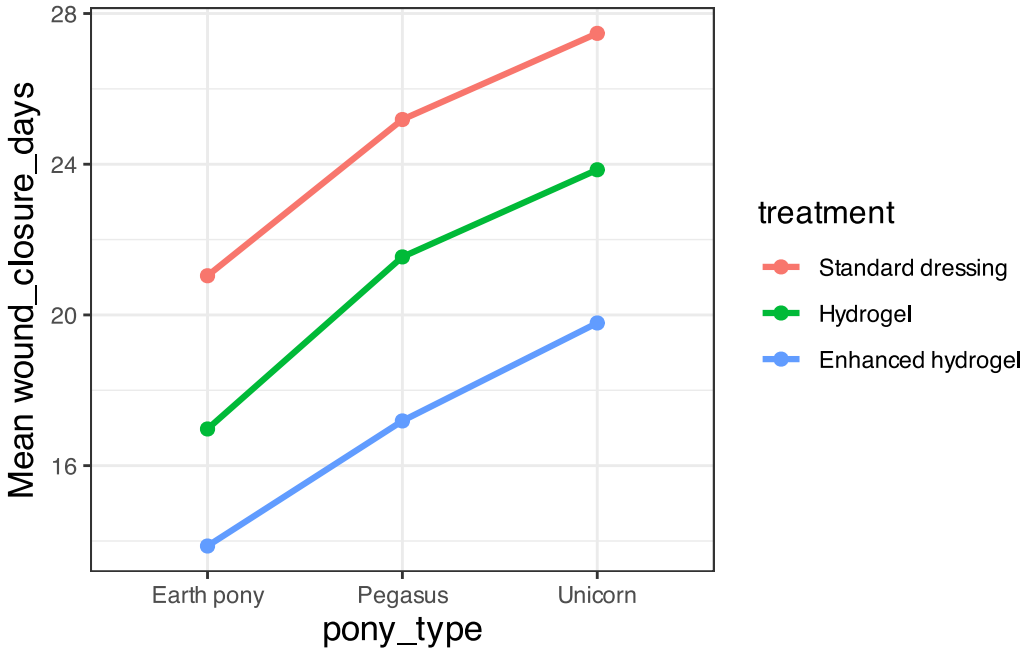
Main effect: pony_type
Comparison             Difference  p-value
──────────────────────────────────────────────
Pegasus vs Earth pony      4.0111  < 0.001  *
Unicorn vs Earth pony      6.4111  < 0.001  *
Unicorn vs Pegasus         2.4000  < 0.001  *
──────────────────────────────────────────────

* p < α = 0.050
```

```
wound_healing |> profile_plot(outcome = wound_closure_days,
                              xaxis = treatment,
                              lines = pony_type)
```




```
wound_healing |> profile_plot(outcome = wound_closure_days,
                              lines = treatment,
                              xaxis = pony_type)
```


Mean wound_closure_days
28
24
20
16
Earth pony
Pegasus
Unicorn
pony_type
treatment
Standard dressing
Hydrogel
Enhanced hydrogel

Let's Practice 4

Rainbow Dash helps the Ponyville clinic compare two fluid treatments after a standardized exercise session. Within each pony type, adult ponies are randomly assigned to Standard fluids or Electrolyte fluids. Time to reach a prespecified hydration target is recorded once for each pony.

```
rehydration |> anovas(outcome = rehydration_time_min,
                      A = treatment,
                      B = pony_type,
                      interaction = FALSE)
```

```
Two-way ANOVA (no interaction)

Source     df         SS         MS        F  p-value
──────────────────────────────────────────────────────
treatment   1  2301.2890  2301.2890  264.232  < 0.001
pony_type   2    36.5502    18.2751    2.098    0.129
Residuals  86   749.0040     8.7093
──────────────────────────────────────────────────────


Hypotheses (treatment):
H₀: All means across levels of treatment are equal
H₁: At least one mean across levels of treatment differs
Test statistic: F(1, 86) = 264.232
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)

Hypotheses (pony_type):
H₀: All means across levels of pony_type are equal
H₁: At least one mean across levels of pony_type differs
Test statistic: F(2, 86) = 2.098
p-value: 0.129
Conclusion: Fail to reject the null hypothesis (p = 0.129 ≥ α = 0.050)
```

```
rehydration |> posthoc(outcome = rehydration_time_min,
                      A = treatment,
                      B = pony_type,
                      interaction = FALSE)
```

```
Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA (rehydration_time_min ~ treatment + pony_type)

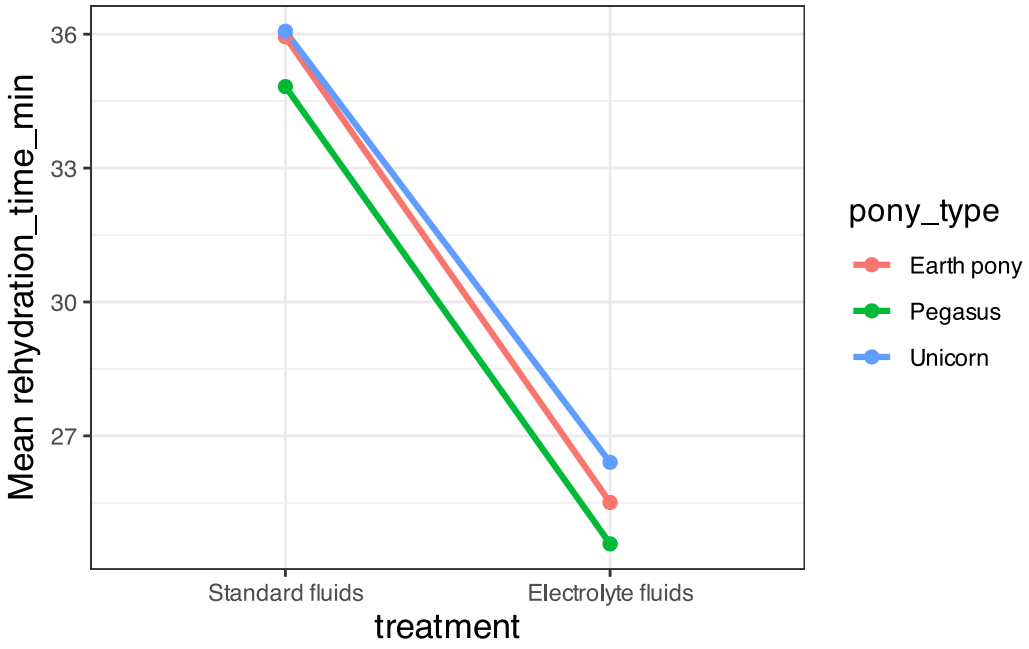
Main effect: treatment
Comparison                          Difference  p-value
```

```
Standard fluids vs Electrolyte fluids     10.1133  < 0.001  *
```

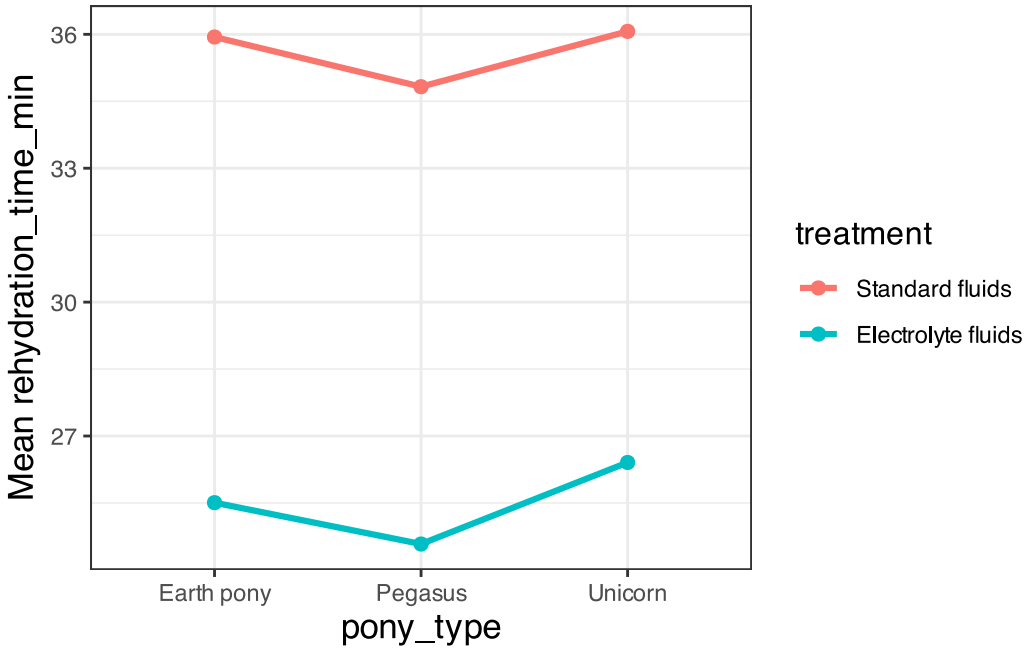
```
Main effect: pony_type
Comparison             Difference  p-value
Pegasus vs Earth pony     -1.0200    0.378
Unicorn vs Earth pony      0.5133    0.779
Unicorn vs Pegasus         1.5333    0.115

* p < α = 0.050
```

```
rehydration |> profile_plot(outcome = rehydration_time_min,
                            xaxis = treatment,
                            lines = pony_type)
```




```
rehydration |> profile_plot(outcome = rehydration_time_min,
                            lines = treatment,
                            xaxis = pony_type)
```


Mean rehydration_time_min
36
33
30
27
Earth pony
Pegasus
Unicorn
pony_type
treatment
Standard fluids
Electrolyte fluids